



Covalent organic frameworks-confined ionic liquids membranes lead to high-efficient CO₂/N₂ separation[☆]

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ABSTRACT

Covalent organic frameworks (COFs) are promising materials for gas separation due to their excellent stability and tunable structure. However, their inherently large pore sizes often limit their application in the separation of small gas molecules, such as CO₂ and N₂. In this study, we regulate the pore size by confining imidazolium-based ionic liquids (ILs) within the TpPa-1 COF using molecular dynamic simulation. A systematic investigation was conducted to explore the effects of IL loading and anion type on the CO₂/N₂ separation. It was found that high IL loading effectively reduces the COF pore size and enhances the dispersion of ILs within COF, resulting in a stratified IL structure, where cations primarily occupy the corners and centers of the COF pores, while anions are staggered near the cations. Due to differences in interactions, CO₂ is adsorbed near the COF pore walls, while N₂ is dispersed in the ILs, which promotes faster diffusion of N₂ compared to CO₂. The size and shape of anions influence their distribution, as well as the interaction and transport of gases. Among the ILs considered, the [Bmim][BF₄]/COF membrane is identified as the optimal candidate for CO₂/N₂ separation.

1. Introduction

Greenhouse gases (GHGs) emitted from thermal power plants and industrial facilities lead to global warming, with carbon dioxide (CO₂) being the primary driver of climate change and the rise in global temperatures [1–6]. Yet, from the point of view of current industrial development, fossil fuels will continue to be the main source of energy. Therefore, it is important to develop effective methods to capture CO₂ from post-combustion emissions, such as those in flue gas [7,8]. Amine-based chemical absorption is currently the most mature technology for capturing CO₂ from flue gas after combustion, but faces severe shortcoming including high energy consumption and the volatilization of organic solvents [9,10]. Therefore, many researchers are now focusing on discovering new solvents or materials to improve CO₂ capture efficiency.

Covalent organic frameworks (COFs) are an emerging class of porous crystalline polymers composed of organic structural units connected by covalent bonds, forming highly ordered and periodic network structures

[11,12]. These materials have attracted significant attention from researchers because of their potential applications in various fields, including gas separation and storage, catalysis, energy storage, and optoelectronics [13–17]. Given their tunable functionalities, well-organized pores, as well as good thermal and chemical stability, the COF membrane, especially those from the Schiff base-related 2D COF family, holds great potential for molecular separation [18,19]. However, the application of COF-based membranes in selective gas separation has been limited, mainly because the nanometer-sized pores of COFs (typically 0.6–10 nm) are much larger than the kinetic diameters of common gas molecules (0.25–0.5 nm) [20].

Ionic liquids (ILs) are a novel class of green media that have attracted great interest in separation, synthesis, catalysis and electrochemistry due to their excellent properties, such as good thermal and chemical stability as well as strong structural designability [21–26]. ILs have shown potential for CO₂ capture [27–33]. However, their industrial application is limited by high viscosity and a relatively small gas–liquid interface, which results in low absorption and desorption rates.

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Composite membranes, fabricated by incorporating ILs to modulate the pore size of COF, combine the advantages of both materials and show great potential for gas separation [34,35]. For example, Jiang et al. [36] developed IL@COF-300/Pebax MMMs using COF-300 modified with [Bmim][TF₂N] by experiments. They found that the addition of [Bmim][TF₂N] reduced the pore size of COF-300 from 1.28 nm to 1.09 nm. Compared to the Pebax membrane, the IL@COF/Pebax membrane exhibited a 209% increase in CO₂ permeability and an 87% improvement in CO₂/CH₄ selectivity. Singh et al. [37] pointed out that the CO₂ adsorption capacity is enhanced by incorporating ILs within COF, with pure COF-300 exhibiting a CO₂ uptake of 3 mmol/g at 1 bar and the COF-300/[Emim][SCN] composite increasing the capacity to 5 mmol/g at 1 bar. Calero et al. [38] employed Monte Carlo simulations and found that COF-confined [Emim][SCN] significantly enhances the selectivity of CO₂/CH₄ at low pressures, with the optimal performance achieved by COF-102 and COF-10 confined with 30% IL. Although COF-confined ILs membranes have demonstrated significant advantages in CO₂ separation, the separation performance strongly depends on the specific combination of ILs and COFs. However, selecting the optimal IL/COF combination from a wide range of ILs and COFs is highly challenging through experimental approaches alone. Therefore, it is essential to investigate the microstructure and interaction mechanisms within these confined membranes to guide the design of high-performance separation materials.

Molecular simulations serve as a powerful tool to complement experimental observations and investigate interaction mechanisms at the molecular level. For example, Zhong et al. [39] investigated the dispersion behavior of ILs in MOF and COF pores using molecular dynamic (MD) and Grand Canonical Monte Carlo (GCMC) simulations. They found that the introduction of [Bmim][SCN] into MOF and COF significantly improves gas separation performance. Furthermore, the dispersion of ILs was demonstrated to play a critical role in separation, with a particularly pronounced effect on CO₂/CH₄ separation. Keskin et al. [40] employed a multiscale computational screening approach, integrating the COSMO-RS method, GCMC simulations, MD simulations, and density functional theory (DFT) calculations, to investigate the adsorption properties and membrane-based CO₂/N₂ separation performance of IL/COF composites. Their results demonstrated that the incorporation of ILs into COFs significantly enhances CO₂/N₂ adsorption selectivity, increasing it from 12 to 26. The carbonyl and fluorine groups in TpPa-1-F2, as well as the aromatic phenyl groups in COF-300, exhibit high affinity for CO₂. Yang et al. [41] found that “Wire-tube” and “Wall-arm” type IL/COF channel structures form in COFs with pore sizes of < 10 Å and ≥ 10 Å, respectively, using a combination of Monte Carlo simulations and machine learning-assisted computational methods.

Although some studies have explored the interaction mechanisms of COF-confined ILs membranes, the relationship between the structural properties of IL/COF membranes and their CO₂ separation performance has not yet been fully elucidated at the molecular level. In particular, the dispersion behavior and kinetic properties of ILs and gases are relatively scarce. In this work, we investigate the microscopic interactions and distributions of imidazolium-based ILs (i.e., [Bmim]⁺ paired with [BF₄][−], [PF₆][−], [SCN][−], [TF₂N][−]) within TpPa-1 COF, as well as their effects on CO₂/N₂ separation performance, using MD simulations. The influence of the IL loading and anion type was systematically examined. The 2D density, radial density, charge density, radial distribution functions, interacting energies, and self-diffusion coefficients were analyzed in detail across different systems, aiming to provide a deeper understanding of the microstructures and properties of COF-confined ILs for CO₂/N₂ separation and to offer valuable insights for designing advanced COF-based IL membranes for CO₂ separation.

2. Model and methods

The structure of TpPa-1 COF was obtained from the CURATED covalent organic frameworks database [42]. By extending the single-cell

model of COF, a larger COF model with dimensions of 39.0681 × 45.112 × 34 Å³ was constructed, comprising 10 layers with an interlayer spacing of 3.4. The chemical structures of [Bmim]⁺, [BF₄][−], [PF₆][−], [SCN][−], and [TF₂N][−] are displayed in Fig. 1(a). Cations and anions were distributed randomly within the COF pores, ensuring an equal number of IL pairs in each pore using the Packmol [43]. Fig. 1(b) shows a snapshot of the initial configuration, featuring 80 pairs of [Bmim][BF₄] confined within the COF.

The effect of confinement on the distribution of ILs was investigated by varying the number of [Bmim][BF₄] pairs to 32, 48, 64, 80, and 96, corresponding to volumetric ratio of 35, 50, 67, 83, and 100 vol%, respectively. The distribution profiles of ILs were compared for different anion types, with the COF pores fully loaded by [Bmim][BF₄], [Bmim][PF₆], [Bmim][SCN], and [Bmim][TF₂N], using 96, 88, 96, and 64 pairs, respectively. To examine the interactions and kinetic properties of CO₂ and N₂ in the confined system, 16 CO₂ and 16 N₂ molecules were introduced into the [Bmim][BF₄]/COF system at 100 vol% loading.

All simulations were performed using the LAMMPS package [44]. The general Amber force field was employed for the ILs [45–48], COF [49–52], CO₂ and N₂ [53]. A cutoff distance of 12 Å was set for the Lennard-Jones (LJ) potential, while the particle–particle particle-mesh (PPPM) method was used to calculate the long-range Coulombic interactions. Periodic boundary conditions were applied in the x, y, and z directions. During the simulations, all COF atoms were frozen, and the ILs were initially heated to 1000 K for 2 ns, followed by quenching to 300 K over an additional 5 ns. Subsequently, a 30 ns NVT simulation at 300 K was conducted for data collection. The Nosé–Hoover algorithm [54] was applied to control the temperature in all simulation systems.

3. Results and discussion

3.1. [Bmim][BF₄] in COFs

3.1.1. 2D density distribution

To investigate the effect of IL loading on the distribution of ILs confined within COF pores, the number density of cations and anions along the YZ plane was computed, as presented in Fig. S1. At lower IL loadings, the ILs exhibit aggregation, with a noticeable movement trend along the Z direction. As the amount of IL increases, the ILs gradually disperse within the COF, indicating that higher IL loading can enhance the dispersion of ILs within the COF pores. The 2D density distribution of ILs in the COF along the XY plane is shown in Fig. 2. When the number of ILs is low, they tend to accumulate near the corners of the pores, which is consistent with the distribution of ILs confined within COFs reported by Zhong et al. [39]. As the IL loading increases, the confined ILs become more uniformly distributed, gradually occupying the entire pore. The cations are mainly distributed at the corners and the center of the COF pores, while the anions are located at the edges of the pores, interspersed among the cations.

In order to illustrate the change in COF pore size after confining IL, the structural diagram of COF with varying loadings of confined IL were analyzed and shown in Fig. 3. It can be observed that the presence of IL inside the pore could reduce the COF pore size, especially at high IL loadings. Furthermore, the pore diameter distributions for the pristine COF, 35 vol% IL/COF, 67 vol% IL/COF, and 100 vol% IL/COF were calculated using Poreblazer software [55] and are displayed in Fig. S2. From Fig. S2, it is evident that the pore diameter for pristine TpPa-1 is approximately 1.5 nm, which corresponds to the internal pore distance excluding the framework. With the loading of 35 vol% IL, the pore diameter remains nearly unchanged. However, when the IL loading is increased to 67 vol%, the pore diameter decreases significantly to below 1 nm, primarily falling within the range of 0.4 to 1 nm. Further increasing the IL loading to 100 vol% results in a substantial reduction of the pore diameter to approximately 0.25–0.4 nm. These results demonstrate that confining IL within COF can effectively reduce the COF pore size, which is consistent with findings reported by other researchers

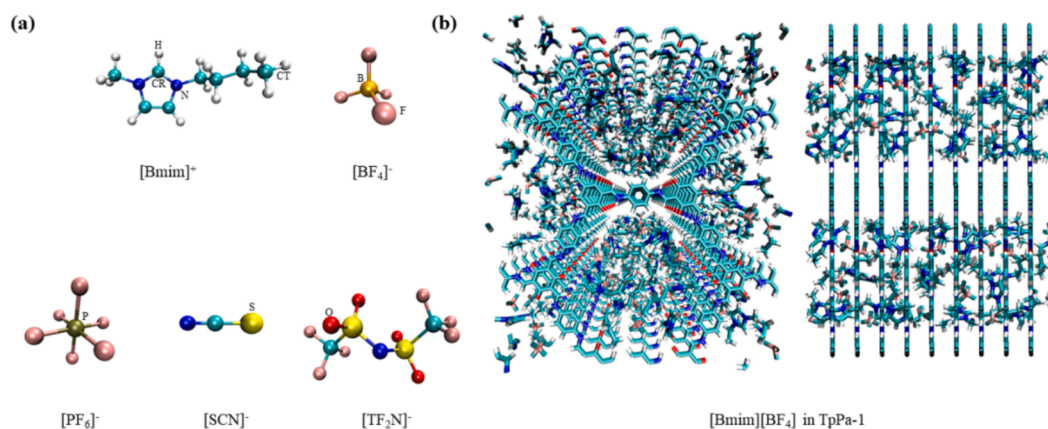


Fig. 1. (a) Structures of cation and anions, (b) the snapshot of $[Bmim][BF_4]$ confined within TpPa-1.

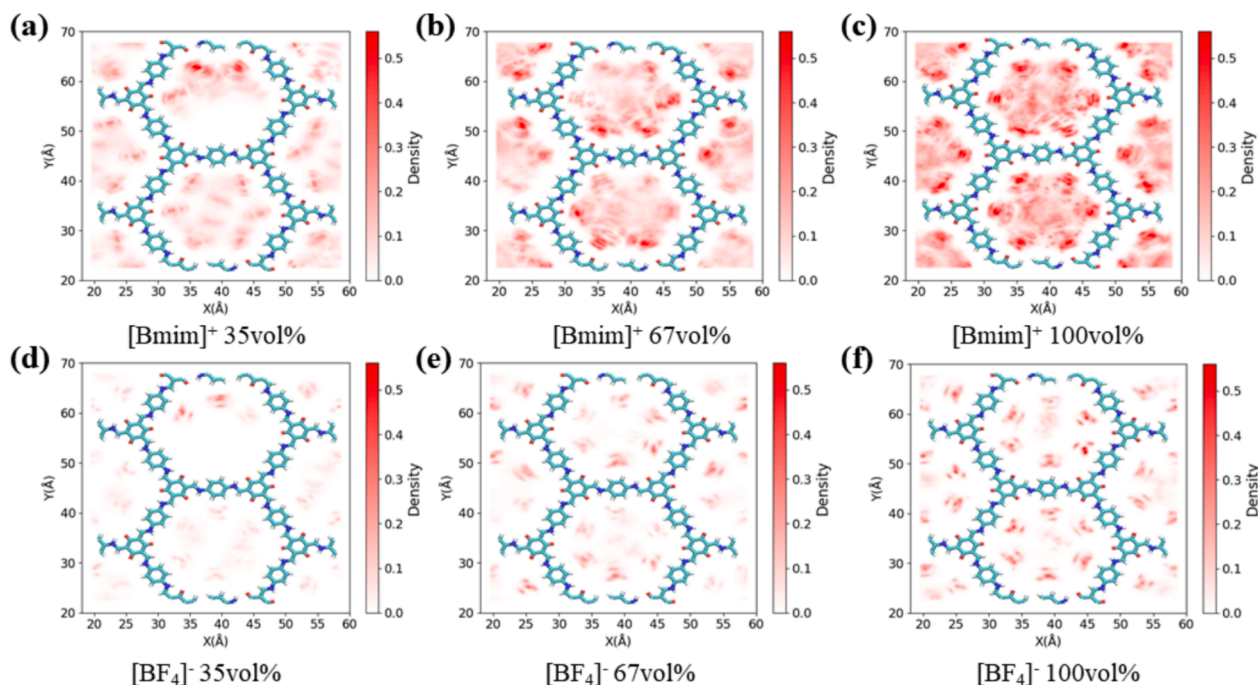


Fig. 2. Density probability of the confined ILs along the XY plane.

[41,56]. Jiang et al.[36] also experimentally demonstrated that the addition of $[Bmim][TF_2N]$ reduced the pore size of COF-300 from 1.28 nm to 1.09 nm.

3.1.2. Radial density

The radial density reflects the preferential radial positions of each species and provides insights into the arrangement of the ILs perpendicular to the pore axis. The number density profiles of anions and cations as a function of the distance from the pore center to the pore surface of the COF were plotted, as shown in Fig. 4. The cation/head refers to the position of the imidazole ring, represented by the C (CR) atom between the two nitrogen atoms in the imidazole ring. The cation/tail refers to the tail of the alkyl side chain, represented by the last C (CT) atom in the side chain. The anion/center denotes the center of the anion, represented by the B atom. The selected atoms are displayed in Fig. 1 (a). For the ILs confined within the nanopores of the COF, a layering behavior is observed in the density profile of anions and cations. The ILs are mainly divided into two layers, which are the central layer and the peripore layer. The studied COF has a pore radius of approximately 0.9 nm, and the cylindrical region with a radius of 0.35 nm is defined as the

central layer (Fig. 4(f)). From Fig. 4, it can be found that when the IL loading is ≤ 50 vol%, the density of cationic heads in the central layer closely matches the density of cationic tails in the peripore layer. Similarly, the density of cationic tails in the central layer matches the density of cationic heads in the peripore layer. The distance between the position corresponding to the highest density layer of cationic tails in the central layer and the position corresponding to the highest density layer of cationic heads around the pore is 0.6 nm, which just is the distance between CR and CT atom. Additionally, in the peripore layer, when the IL loading is ≤ 50 vol%, the highest value of number density of the cation head is significantly closer to the COF pore wall compared to the center of the anion. However, when the IL loading is > 50 vol%, the highest value of number density at the cation head is located at the same position as that of the anion center. This suggests that at lower IL loadings, the pore space is larger, and the cations in ILs are closer to the COF pore periphery, likely due to the strong interaction between the cations and the COF. As the number of ILs increases, the ILs occupy more of the COF pore space, restricting the distribution of anions and cations. However, a delamination phenomenon becomes more pronounced between the central layer and the peripore layer. The extent of the peripore

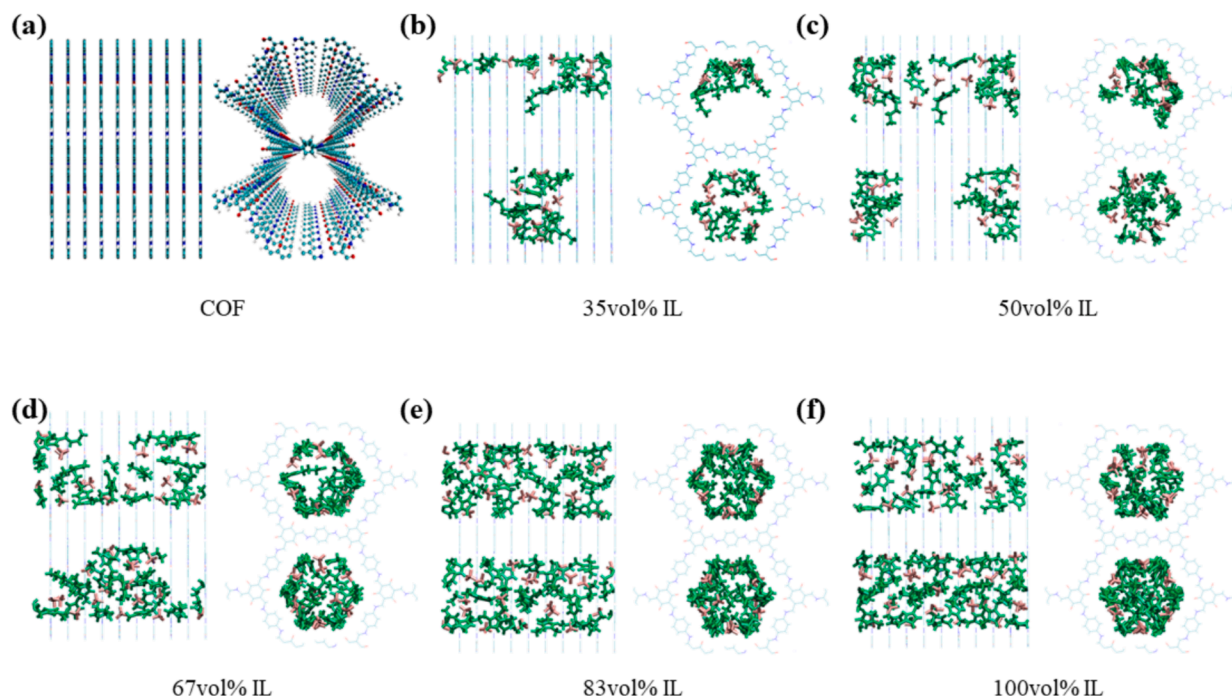


Fig. 3. The structural diagram of COF with varying loadings of confined IL.

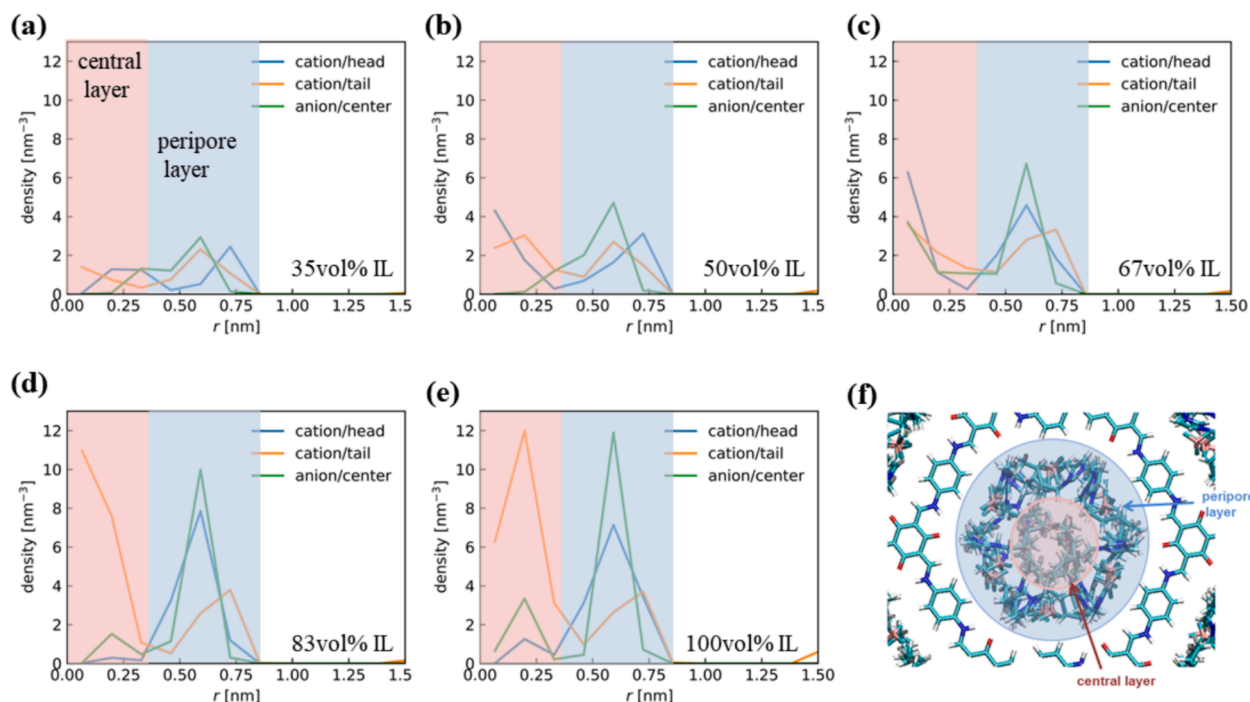


Fig. 4. (a)-(e) Radial number density distribution of cations and anions in pore, (f) Diagram of the ILs layering phenomenon.

and central layers remains unchanged with increasing IL loading.

The distribution of cation heads and tails was further analyzed through the cation orientation distribution, as plotted in Fig. S3 and Fig. S4. The definition of cationic orientation and the calculation of $\cos\theta$ are provided in the supporting material. The value of $\cos\theta > 0$ indicates that the imidazole ring of the cation points toward the pore center, while its side chains point toward the periphery of the pore. Conversely, $\cos\theta < 0$ implies that the imidazole ring points toward the periphery of the pore, with the side chains directed toward the center. As shown in

Fig. S4, when the IL loading is ≤ 67 vol%, the value of $\cos\theta$ at the pore center is positive, while the value of $\cos\theta$ at the pore periphery tends toward -1 , suggesting that the cations are oriented as depicted in Fig. S4 (d). Both orientations, with the imidazole ring facing either the pore periphery or the pore center, are present. When the IL loading exceeds 67 vol%, the number of cation/tail in the middle layer is the largest, with most cationic heads located in the peripore layer. This implies that, at higher IL loadings, the imidazole ring points toward the periphery of the pore, while the side chains point toward the pore center, as shown in

Fig. S4 (g). These results suggest that the amount of ILs significantly influences the orientation of anions and cations.

Fig. S5 depicts the charge density profiles for cations (left), anions (middle) and ILs (right). The positively charged cations are distributed near the oxygen (O) atoms in the COF, which carry a negative charge of -0.57 . It can be deduced that electrostatic interactions between the positively charged cations and the negatively charged O atoms in the COF drive the cations toward the corners of the pores. In contrast, the anions are distributed around the cations, interspersed with them due to strong electrostatic interactions. These findings reveal the significant role of electrostatic forces in determining the spatial arrangement of ILs within the COF pores.

3.1.3. Interaction

Radial distribution functions (RDFs) describe the probability of finding other atoms or positions around a central atom or position at a given distance. The N atom (the nitrogen atom in the side chain of the imidazole ring) and the B atom represent cations and anions, respectively. The site-to-site RDFs between the cations and anions in systems with different numbers of ILs are shown in Fig. 5. As the number of ILs increases, the position of the first peak in the cation–anion RDF shifts to larger distance, indicating that cations and anions move further apart from each other. Simultaneously, the height of the first peak decreases with increasing IL numbers, although it remains higher than that of bulk IL, suggesting that the ILs become more spatially dispersed. This observation is consistent with the microstructures shown in Fig. 2.

Furthermore, the interactions between the anions and cations in the pure IL system, as well as those between confined ILs and the COF, were calculated and are shown in Fig. 5(b) and Table S1. As displayed in Table S1, the interaction energy between the COF and cations is slightly higher than that between COF and anions. However, the interaction energy between cations and anions is much higher than that between the COF and ILs, being three times greater than the COF–ILs interaction energy. It is also found from Fig. 5(b) that the anions–cations interaction energy in bulk IL is greater than that in the confined system. Combined with the above analyses, it is evident that the confined $[\text{Bmim}]^+$ interacts with the COF through electrostatic interactions, leading to a decrease in the interaction energy between the anions and the cations. These results are consistent with the findings of Zhang et al. [57].

3.2. Different anions-based ILs in COF

3.2.1. 2D density distribution

It is well known that the type of anion plays an important role in

determining CO_2 separation [58]. Therefore, the effect of different anion types on the distribution of ILs within COF was explored. ILs at a loading of 100 vol% were chosen for further examination. The ILs examined are $[\text{Bmim}][\text{BF}_4]$, $[\text{Bmim}][\text{PF}_6]$, $[\text{Bmim}][\text{SCN}]$, and $[\text{Bmim}][\text{TF}_2\text{N}]$. As shown in Fig. 6, regardless of the anion type, the cations are always distributed at the corners and center of the COF pores, while the anions occupy the edge regions of the pores, interspersed among the cations. The specific distribution patterns of anions are influenced by their size and structure. When the anion is a small chain-like structure (e.g., $[\text{SCN}]^-$), it can fit more closely into the corners of the COF pore, resulting in a tighter arrangement. When the anion is a larger chain-like structure (e.g., $[\text{TF}_2\text{N}]^-$), it is more dispersed around the pore periphery but remains adjacent to the cations, leading to a more distinct separation between the peripheral cation layer and the central pore layer. For anions with a symmetrical spherical structure (e.g., $[\text{BF}_4]^-$, $[\text{PF}_6]^-$), they exhibit a relatively uniform distribution along the pore periphery. The larger the volume of anion (e.g., $[\text{PF}_6]^-$), the more pronounced the spacing between the peripore layer and the central cation layer.

3.2.2. Radial density analysis

The radial density profiles of anions and cations with different anion types are plotted, as shown in Fig. 7. The P, C, and N atoms were selected as the central atoms of $[\text{PF}_6]^-$, $[\text{SCN}]^-$ and $[\text{TF}_2\text{N}]^-$, respectively. From Fig. 7, it can be observed that the distribution of different ILs exhibits strong stratification, divided into two layers: the central layer and the peripore layer, similar to the distribution of $[\text{Bmim}][\text{BF}_4]$ shown in Fig. 4(e). In addition, the extent of the central layer remains consistent across different anions, with significantly more cations than anions present in the central layer. The position of anions is related to their size and shape [59]. For instance, $[\text{SCN}]^-$, being the smallest anion, is located closest to the pore periphery, with almost no $[\text{SCN}]^-$ found in the central layer, consistent with the observations in Fig. 6(e). On the other hand, $[\text{TF}_2\text{N}]^-$, which has a larger and more chain-like structure, is located farthest from the pore periphery, and almost no $[\text{TF}_2\text{N}]^-$ is present in the central layer. However, for $[\text{BF}_4]^-$ and $[\text{PF}_6]^-$, which have symmetrical, spherical shapes, are present in small amounts in the central layer. Furthermore, the charge density distributions of various anions and cations were also calculated and are shown in Fig. S6. The charge distribution of the anions is closely related to their geometric structures. For anions with symmetric structures, the charge distribution within the pore is also symmetric. However, for anions with long-chain structures, the charge distribution exhibits asymmetry.

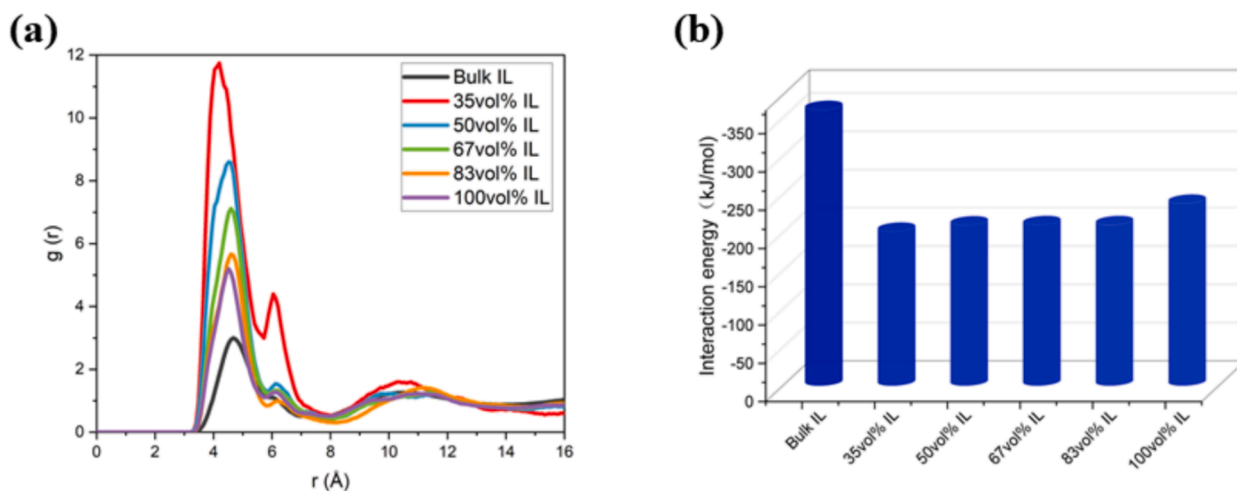


Fig. 5. (a) The site-to-site RDFs between the cations and the anions in the systems with different IL numbers, (b) Interaction energies of cations–anions in the bulk IL and ILs/COF systems.

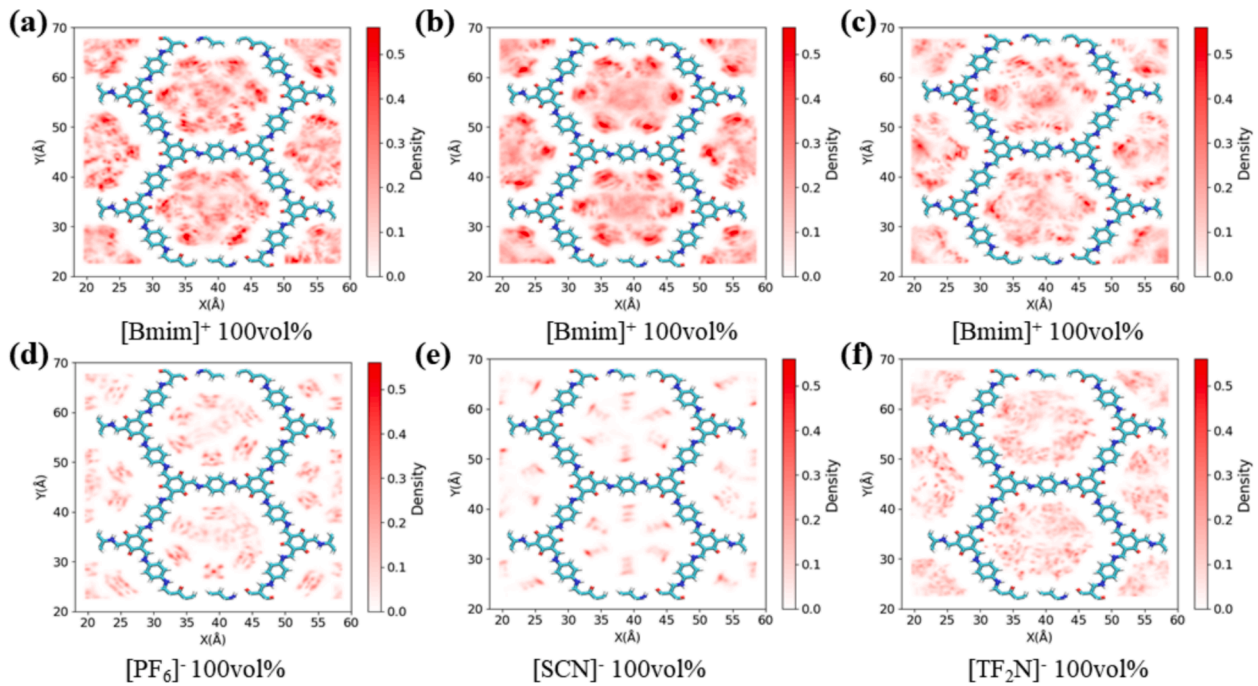


Fig. 6. Density probability of the confined ILs along XY plane with different anions.

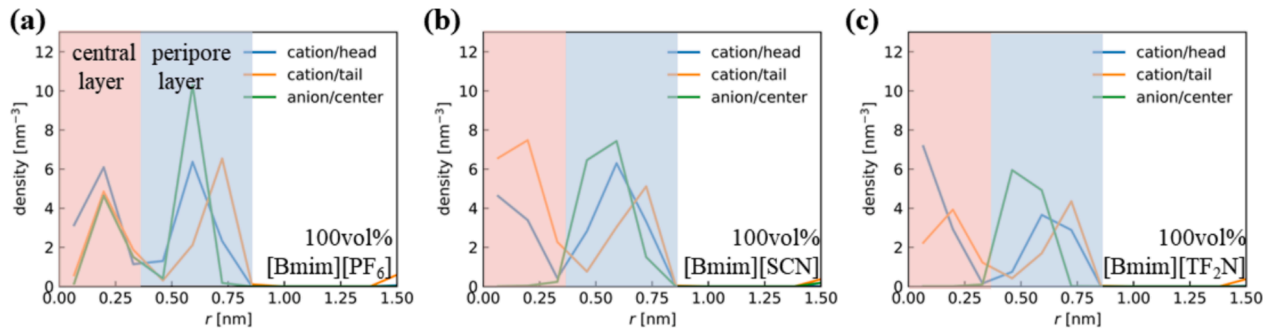


Fig. 7. Radial number density distribution of cations and anions in pore.

3.2.3. Transport property

To explore the motion of ILs and gas inside the pore, the diffusion coefficients were calculated from the slope of the mean square displacement (MSD) of the ILs and gas versus time. The self-diffusion coefficients of ILs, CO₂ and N₂ in all simulated systems were derived from the Einstein relation [60],

$$D = \frac{1}{6N} \lim_{t \rightarrow \infty} \frac{d}{dt} \left(\sum_{i=1}^N [r_i(t) - r_i(0)]^2 \right)$$

where N is the total number of molecules, and $r_i(0)$ and $r_i(t)$ are the original and final positions, and the amount in the angle brackets is the tethered average of the MSDs. The self-diffusion coefficients of ILs in both bulk and confined systems were calculated, as shown in Table 1. In bulk systems, the self-diffusion coefficients of ILs follow the order [Bmim][SCN] > [Bmim][TF₂N] > [Bmim][BF₄] > [Bmim][PF₆], which aligns with the viscosity trend of pure ILs. Additionally, the self-diffusion coefficients of [Bmim]⁺ and [PF₆]⁻ agree well with the molecular dynamics simulation results reported by Zhang et al. [57]. These findings collectively confirm the reliability of our results. Furthermore, by comparing the self-diffusion coefficients of ILs in bulk and confined systems, it is evident that the self-diffusion coefficients of ILs in the COF are significantly lower than those in bulk ILs. For example, the self-

Table 1

Self-diffusion coefficients of cations and anions in bulk and confined ILs.

Self-Diffusion Coefficients	Bulk ILs (1×10^{-12} m ² /s)			Confined ILs (1×10^{-13} m ² /s)		
	Cation	Anion	ILs	Cation	Anion	ILs
[Bmim][BF ₄]	1.73 ± 0.0037	0.64 ± 0.0027	1.19 ± 0.0064	0.98 ± 0.1002	0.34 ± 0.1764	0.66 ± 0.2766
[Bmim][PF ₆]	1.32 ± 0.0041	0.53 ± 0.0035	0.93 ± 0.0076	0.42 ± 0.0388	0.52 ± 0.0364	0.47 ± 0.0752
[Bmim][SCN]	1.93 ± 0.0027	1.45 ± 0.0061	1.74 ± 0.0088	1.71 ± 0.0556	2.13 ± 0.0498	1.92 ± 0.1054
[Bmim][TF ₂ N]	2.27 ± 0.0617	1.03 ± 0.0155	1.65 ± 0.0772	4.40 ± 0.0459	4.64 ± 0.0421	4.52 ± 0.0880

diffusion coefficient of pure [Bmim][SCN] is 1.74×10^{-12} m²/s, while in the COF, it is 1.92×10^{-13} m²/s. The self-diffusion coefficient of ILs in confined system is an order of magnitude lower than that in bulk, suggesting that confinement within the COF significantly reduces the diffusion of ILs. In the confined system, [Bmim][TF₂N] exhibits the fastest diffusion at 4.52×10^{-13} m²/s, while [Bmim][PF₆] shows the slowest diffusion.

3.3. COF/ILs-gas

3.3.1. Distribution of gas

In order to understand the gas distribution within the pore, the 2D density distribution of CO₂, N₂ and ILs were analyzed. Using [Bmim][BF₄] as an example, the 2D density diagrams along the XY plane were compared, as shown in Fig. 8 and Fig. S7. Fig. S7 reveals that the density distribution of [Bmim]⁺ and [BF₄][−] in the COF/ILs-gas system are nearly identical to those in the COF/ILs system, which suggests that the presence of gas does not affect the distribution of ILs. Fig. 8 (a) and (c) depict the position distribution of CO₂ and N₂ within the pore along the XY plane, and (b) and (d) present the actual motion trajectories of CO₂ and N₂ inside a single pore during simulation time. In order to clearly illustrate the gas distribution, the positions of ILs are not shown. As displayed in Fig. 8 (a) and (c), CO₂ predominantly localizes near the corners of COF pore wall, close to C=O and N–H groups, whereas N₂ is more evenly dispersed within the pore. Experimental studies have confirmed that TpPa-1 exhibits strong adsorption capacity for CO₂ [61], which is attributed to the electrostatic interactions between the C=O and N–H groups of TpPa-1 and CO₂ [62]. From the trajectories of CO₂ and N₂ within the COF, as presented in Fig. 8 (b) and (d), it is observed that CO₂ tends to remain localized near the COF pore and exhibits minimal movement. In contrast, N₂ displays more pronounced mobility and can be found at various positions within the pore. These results indicate that COF has a strong affinity for CO₂, which hinders its transport. Similar phenomena have also been reported for CO₂ in other COF membranes, such as COF-300 [40], PVA@TpPa-SO₃H [63], COFMs [64], and LA-α-CD-in-TpPa-1 membranes [65].

3.3.2. Interaction energy

In order to further investigate the interaction between gas and COF/ILs after confinement, the interaction energies between gas and COF, as well as between gas and ILs, were compared, as shown in Fig. 9(a). For the four ILs studied, the interaction energies between CO₂ and COF, as well as between CO₂ and ILs, are greater than those between N₂ and COF or between N₂ and ILs. This is attributed to the fact that N₂ does not react easily with other materials. In addition, the interaction energy between CO₂ and COF is much larger than that between CO₂ and ILs, while the

interaction energy between N₂ and ILs is higher than that between N₂ and COF. This suggests that the strong interaction between COF and CO₂ causes CO₂ to be confined around the COF pore, preventing its movement. Whereas, the strong interaction between ILs and N₂ attracts N₂ into the pore. This explains why CO₂ remains localized near the pore corners, while N₂ is more dispersed within the pore. Moreover, since the interaction energy between CO₂ and COF is the highest, the interaction energies between CO₂ and COF for the four different confined ILs were compared. As displayed in Fig. 9(a), the interaction energy between CO₂ and COF follows the order of [Bmim][TF₂N] > [Bmim][PF₆] > [Bmim][BF₄] > [Bmim][SCN]. In contrast, for N₂, the interaction with ILs is stronger. Therefore, the interaction energies between N₂ and ILs were further compared for the four different confined ILs, with the sequence of N₂-[Bmim][SCN] > N₂-[Bmim][BF₄] > N₂-[Bmim][TF₂N] > N₂-[Bmim][PF₆]. Therefore, from the perspective of interaction energy, COF-confined IL membranes with strong interactions between the COF and CO₂, but simultaneously strong interactions between the ILs and N₂, are required for the effective separation of CO₂ and N₂. As shown in Fig. 9(a), among the four ILs studied, [Bmim][TF₂N] and [Bmim][BF₄] are suitable conditions.

3.3.3. Diffusion of gas

Furthermore, the self-diffusion coefficients of CO₂ and N₂ were calculated. Taking COF-confined [Bmim][SCN] as an example, the MSD values as a function of simulation time for CO₂ and N₂ in COF-confined [Bmim][SCN] membrane are shown in Fig. S8. The self-diffusion coefficients of CO₂ and N₂ were calculated by linearly fitting the slope in the region from 15 to 25 ns and the values are listed in Fig. 9 (b). From Fig. 9(b), it is clear that the diffusion coefficient of N₂ is always higher than that of CO₂, which is consistent with the energy results. The faster diffusion of N₂ compared to CO₂ has also been observed in COF-300/[BMPyrr][DCA], COF-300, TpPa-1-F2 and COF-300/Pebax(7) [36,40] membranes through both experimental and simulation studies. Additionally, the diffusion coefficients of CO₂ in the four confined ILs follow the order of [Bmim][TF₂N] < [Bmim][PF₆] < [Bmim][BF₄] < [Bmim][SCN], which is inversely related to the interaction energy between CO₂ and COF. The larger the interaction energy between CO₂ and COF, the more CO₂ is attracted around COF, making it less likely to diffuse.

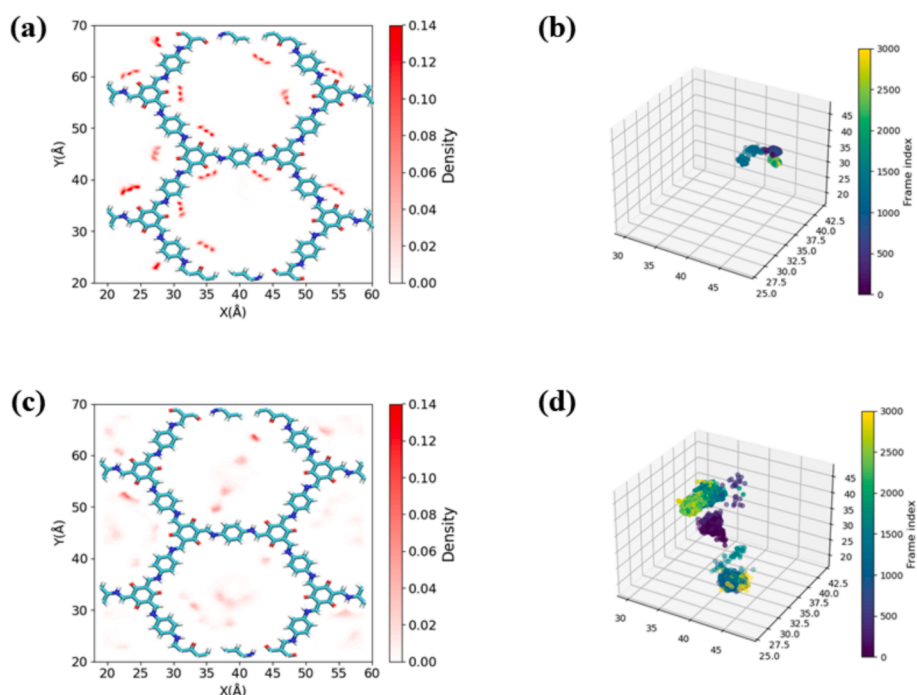


Fig. 8. 2D density distribution of (a) CO₂ and (c) N₂ in pores, trajectory of (b) CO₂ and (d) N₂ in the pore with time.

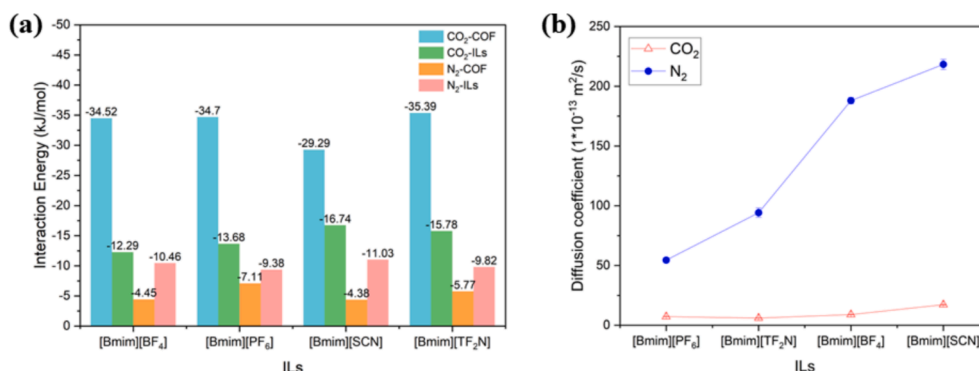


Fig. 9. (a) Interaction energy between CO₂ and COF, CO₂ and ILs, N₂ and COF, N₂ and ILs, (b) Self-diffusion coefficients of CO₂ and N₂.

Therefore, in the composite membrane formed by COF-confined ILs, CO₂ is adsorbed on the COF pore walls due to the strong interactions between CO₂ and COF. This observation is consistent with the findings reported by Zhao et al. [62].

For N₂, the diffusion coefficient in the four confined ILs is in the sequence of [Bmim][SCN] > [Bmim][BF₄] > [Bmim][TF₂N] > [Bmim][PF₆], which is positively correlated with the interaction between N₂ and ILs. The greater the interaction of N₂ with ILs, the greater the solubility of N₂ in ILs, facilitating its diffusion. Moreover, the diffusion selectivity of N₂/CO₂ in the four confined ILs was analyzed. The diffusion selectivity values for [Bmim][BF₄], [Bmim][TF₂N], [Bmim][SCN] and [Bmim][PF₆] are 21.05, 15.67, 12.69 and 7.54, respectively. These results indicate that the diffusion selectivity of N₂/CO₂ is the highest in [Bmim][BF₄]. In addition, the comparison between the self-diffusion coefficients of CO₂ and N₂ for individual gas molecules and those in gas mixtures is presented in Fig. S9. It can be observed that the self-diffusion coefficients of CO₂ and N₂ in the gas mixtures are higher than those of the individual gas molecules, which aligns with the findings reported by Wang et al. [63]. The [Bmim][BF₄]/COF membrane still exhibits the highest diffusion selectivity for N₂ and CO₂ in the gas mixture. Therefore, by comprehensively considering both interaction energy and diffusion selectivity, among the four ILs studied, the [Bmim][BF₄]/COF membrane is the most suitable for the separation of CO₂/N₂.

4. Conclusions

In this work, the ILs were confined within COF to adjust the pore size for CO₂/N₂ separation, and the microstructures and interactions were systematically studied by molecular dynamics simulations. The effects of IL loading and anion type were investigated. We found that confining 100 vol% ILs within TpPa-1 significantly reduces the pore size from 1.5 nm to 0.25–0.4 nm. As the IL loading increases, the anions and cations transition from aggregating in the corners of the COF pores to uniformly dispersing within the pores, accompanied by a noticeable layering phenomenon. The cations are attracted to the O atoms on the COF through electrostatic interactions, causing them to distribute around the COF pores. Meanwhile, the anions are interspersed with cations around the pores due to the stronger interactions between anions and cations. The interaction between COF and cations is slightly stronger than that between COF and anions. The IL loading influences the orientation of the cations. When IL loading is lower than 67 vol%, both orientations of the imidazole ring toward the pore periphery and center are observed. However, when IL loading exceeds 67 vol%, the imidazole ring orients towards the pore periphery. After confinement, the interaction between anions and cations is weaker than in bulk ILs. For the different anion types, their size and shape affect their distribution within the pores, and the self-diffusion coefficients of anions and cations are significantly reduced after confinement. The interaction energy between CO₂ and COF/ILs is greater than that between N₂ and COF/ILs, and the self-diffusion coefficient of CO₂ is much smaller than that of N₂. The

interaction energy between CO₂ and COF is larger than that between CO₂ and ILs, and the self-diffusion coefficient of CO₂ is negatively correlated with the interaction energy between CO₂ and COF. Conversely, the interaction energy between N₂ and COF is smaller than that between N₂ and ILs, and the self-diffusion coefficient of N₂ is positively correlated with the interaction energy between N₂ and ILs. The [Bmim][BF₄]/COF membrane is recommended for CO₂/N₂ separation due to its stronger interactions between the gas molecules and the COF/IL framework, as well as its superior diffusion selectivity. The excellent CO₂/N₂ separation efficiency is attributed to the synergistic interaction of COF and ILs, where the COF strongly attracts CO₂, hindering its transport, while the ILs regulate pore size and facilitate N₂ diffusion. Therefore, for COF-confined IL membranes designed for CO₂/N₂ separation, it is essential to select COFs with polar functional groups (e.g., C=O and N–H) and ILs that exhibit strong interactions with N₂. This work provides valuable insights for designing membranes with outstanding separation performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seppur.2025.132933>.

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